CUSTOMER SENTIMENT AND INTENT ANALYSIS PLATFORM FOR BUSINESS INTELLIGENCE

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

DSA0307 – NATURAL LANGUAGE PROCESSING

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

C. Jaswanth(192424334)
M. Phanith Chandu(192425238)
S. Mahammed Nayeem(192425294)
B. Ganga Niranjan (192424041)
S. Tushar Krishna Sai (192424406)

Under the Supervision of

DR. K. ANBAZHAGAN
SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

SEPTEMBER 2026

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **"Customer Sentiment And Intent Analysis Platform For Business Intelligence"** has been carried out by **C. Jaswanth (192424334), M. Phanith Chandu (192425238), S. Mahammed Nayeem (192425294) , B. Ganga Niranjan (192424041), S. Tushar Krishna Sai (192424406)** under the supervision of **Dr. K. Anbazhagan** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech in Computer science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr Devi T,

Professor,
Deprtment of CSE
SIMATS Engineering,
SIMATS, Chennai – 602105.

COURSE FACULTY

Dr. K. Anbazhagan

Professor
Deprtment of CSE
SIMATS Engineering,
SIMATS, Chennai – 602105.

DECLARATION

We, **C. Jaswanth (192424334), M. Phanith Chandu (192425238), S. Mahammed Nayeem (192425294), B. Ganga Niranjan (192424041), S. Tushar Krishna Sai (192424406) of the** Computer Science And Engineering , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **"Customer Sentiment And Intent Analysis Platform For Business Intelligence"** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date:08-09-2026

Signature of the Students with Names

C. Jaswanth(192424334)
M. Phanith Chandu(192425238)
S. Mahammed Nayeem(192425294)
B. Ganga Niranjan (192424041)
S. Tushar Krishna Sai (192424406)

Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Contribution (%)
C. Jaswanth (192424334)	Database Design	Schema Design	Database Testing	Design Section	20%
M. Phanith Chandu (192425238)	SQL Development	SQL Queries	Query Testing	Implementation Section	20%
S. Mahammed Nayeem (192425294)	Indexing	B+ Tree Indexing	Performance Analysis	Indexing Section	20%
B. Ganga Niranjan (192424041)	System Testing	Module Integration	Functional Testing	Testing Section	20%
S. Tushar Krishna Sai (192424406)	Documentation	System Refinement	Result Analysis	Conclusion & Report	20%

Total		100%

ABSTRACT

The Customer Sentiment and Intent Analysis Platform for Business Intelligence addresses the engineering problem of extracting meaningful insights from large volumes of customer feedback, reviews, comments, and textual interactions. Traditional methods of analyzing customer opinions manually are time-consuming, subjective, and difficult to scale. Businesses therefore require an automated platform that can understand customer emotions and identify the intentions behind their feedback to support faster and more accurate decision-making.

The proposed platform uses Natural Language Processing (NLP) and machine learning techniques to analyze customer-generated text. It identifies the sentiment of each customer interaction as positive, negative, or neutral and determines the underlying intent, such as complaint, inquiry, appreciation, feedback, or purchase-related interest. The analyzed information is transformed into useful business intelligence that can help organizations understand customer expectations and identify areas requiring improvement.

The methodology involves collecting customer feedback, preprocessing the textual data through cleaning and normalization, performing tokenization and feature extraction, and applying sentiment and intent classification techniques. The platform then analyzes the classified data and generates summarized insights through charts, dashboards, and reports. Performance is evaluated using suitable metrics such as accuracy, precision, recall, and F1-score.

The major results demonstrate that automated sentiment and intent analysis can efficiently classify customer opinions and identify recurring customer requirements. The platform helps reveal sentiment trends, common customer concerns, and frequently expressed intentions, reducing the effort required for manual analysis. These insights can support organizations in improving products, services, customer support, and overall business strategies.

The project concludes that integrating NLP-based sentiment and intent analysis with business intelligence provides an effective solution for converting unstructured customer feedback into actionable insights. The proposed platform enables organizations to understand customer behavior, monitor satisfaction trends, identify critical issues, and make data-driven business decisions.

Keywords

Customer Sentiment, Intent Analysis, Natural Language Processing, Business Intelligence, Machine Learning, Text Classification

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1
INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

In the modern digital business environment, customers interact with organizations through online reviews, feedback forms, social media, emails, chat systems, and customer support platforms. These interactions generate large amounts of unstructured textual data. Although this data contains valuable information about customer satisfaction, expectations, complaints, and purchasing interests, manually analyzing it is time-consuming and difficult.

The **Customer Sentiment and Intent Analysis Platform for Business Intelligence** is designed to automatically process and analyze customer-generated text using **Natural Language Processing (NLP)** and machine learning techniques. Sentiment analysis identifies whether a customer's opinion is positive, negative, or neutral, while intent analysis determines the purpose behind the customer's message, such as complaint, inquiry, appreciation, feedback, or purchase interest.

The platform converts unstructured customer text into structured information that can be presented through dashboards, charts, and reports. These insights enable businesses to understand customer behavior, identify common problems, monitor satisfaction levels, and support data-driven decision-making.

1.2 Need for the Project

The project is needed because businesses receive a large volume of customer feedback that cannot be efficiently analyzed through manual methods.

- **Problem Relevance:** Manual analysis of customer feedback requires significant time and effort.
- **Affected Users:** Businesses, customer service teams, marketing teams, product managers, and decision-makers are affected.
- **Existing Limitations:** Traditional methods may be slow, subjective, difficult to scale, and unable to identify sentiment and intent consistently.
- **Business Need:** Organizations need timely insights to improve customer satisfaction and service quality.
- **Engineering Significance:** NLP and machine learning can automate text analysis and convert unstructured data into useful business intelligence.
- **Decision Support:** The system helps identify customer concerns, positive responses, and emerging trends for better decision-making.

1.3 Problem Statement

Businesses generate large volumes of customer feedback through multiple digital channels, making manual analysis inefficient and difficult to scale. The engineering problem is to design and develop an automated platform capable of processing unstructured customer text, accurately identifying **sentiment and intent**, handling variations in language and text quality, and presenting the results as meaningful business intelligence.

The system must balance multiple requirements, including **classification accuracy, processing efficiency, scalability, usability, and reliable interpretation of customer feedback**. It should handle noisy or inconsistent text while producing understandable results for business users. The solution should also operate within realistic constraints such as limited datasets, computational resources, classification errors, and variations in customer language.

1.4 Project Aim

The aim of this project is to **design and develop an NLP-based Customer Sentiment and Intent Analysis Platform that converts customer feedback into actionable business intelligence for effective decision-making.**

1.5 Project Objectives

1. To design a platform for collecting and processing customer feedback.
2. To develop an NLP-based text preprocessing pipeline for customer-generated data.
3. To model customer sentiment into positive, negative, and neutral categories.
4. To implement intent classification for identifying customer requirements and concerns.
5. To develop dashboards for visualizing sentiment and intent analysis results.
6. To test the system using different customer feedback samples.
7. To evaluate classification performance using accuracy, precision, recall, and F1-score.
8. To generate actionable insights that support business decision-making.

1.6 Scope

Included

- Collection and processing of customer feedback.
- Text cleaning and preprocessing.
- Tokenization and feature extraction.
- Sentiment classification.
- Customer intent classification.
- Data analysis and visualization.
- Dashboard and report generation.
- Performance evaluation using classification metrics.

Excluded

- Voice-based sentiment analysis.
- Real-time facial or emotion recognition.
- Direct modification of business systems based on predictions.
- Guaranteed detection of sarcasm, slang, or complex emotions.
- Analysis of private customer information not provided for the system.

Assumptions

- Customer feedback is available in text format.
- The input data is sufficiently relevant for sentiment and intent analysis.
- The training and testing datasets contain representative examples.
- The selected NLP and machine learning models are suitable for the available dataset.

Boundary Conditions

- The system primarily analyzes text-based customer feedback.

- Prediction quality depends on the quality and diversity of the dataset.
- The platform provides analytical recommendations rather than autonomous business decisions.
- Classification results may contain errors for ambiguous or context-dependent statements.

1.7 Expected Engineering Outcome

The expected engineering outcome is a **functional prototype of a Customer Sentiment and Intent Analysis Platform for Business Intelligence**. The system will provide an integrated process for collecting customer text, preprocessing the data, classifying sentiment and intent, analyzing results, and presenting insights through visual dashboards.

The expected solution includes:

- **System:** NLP-based customer feedback analysis platform.
- **Product:** Business intelligence dashboard for customer insights.
- **Process:** Automated customer feedback processing and classification.
- **Algorithm:** Sentiment and intent classification algorithms.
- **Model:** Machine learning/NLP classification model.
- **Prototype:** Working application demonstrating automated analysis and visualization.
- **Output:** Sentiment trends, intent distribution, customer concerns, and actionable business insights.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

A review of existing research, technologies, and applications related to customer sentiment analysis, intent detection, Natural Language Processing (NLP), machine learning, and business intelligence was conducted. Relevant academic studies and existing systems were examined to understand the methods currently used for processing customer feedback.

The review considered different approaches based on their accuracy, processing speed, scalability, implementation complexity, and usefulness for business decision-making. Traditional NLP techniques, machine learning algorithms, deep learning approaches, and visualization methods were compared. The limitations identified during the review were used to determine the requirements of the proposed platform.

2.2 Review of Existing Work

2.2.1 Customer Sentiment Analysis

Customer sentiment analysis is an important application of NLP that determines the emotional polarity of customer feedback. Most basic systems classify feedback as positive, negative, or neutral. Businesses use sentiment information to measure customer satisfaction and identify negative experiences.

Traditional sentiment analysis methods often use predefined dictionaries or machine learning models. Although these methods are simple and efficient, they may have difficulty understanding sarcasm, mixed opinions, informal language, and context-dependent expressions.

2.2.2 Customer Intent Analysis

Intent analysis focuses on understanding the purpose of a customer's message. For example, a customer may submit a message to make an inquiry, report a problem, request a product, provide feedback, or express appreciation.

Intent classification can help businesses route customer requests to the appropriate department and identify frequently occurring customer requirements. However, accurately identifying intent can be difficult when a single message contains multiple topics or unclear information.

2.2.3 NLP Text Preprocessing

Raw customer feedback usually contains unnecessary words, punctuation, spelling variations, and other noise. NLP preprocessing techniques are therefore applied before classification.

Common preprocessing steps include:

- Text cleaning
- Lowercase conversion
- Tokenization
- Stop-word removal
- Stemming
- Lemmatization
- Feature extraction

Effective preprocessing improves the quality of input data and can increase classification performance.

2.2.4 Machine Learning Approaches

Machine learning algorithms are widely used to classify customer feedback. Naive Bayes, Logistic Regression, Support Vector Machine, and Random Forest are commonly considered for text classification tasks.

These algorithms can provide efficient results with suitable training data. However, their performance depends on the quality of the dataset, feature representation, and training process. Traditional models may also have limitations when processing complex language and understanding deeper context.

2.2.5 Deep Learning Approaches

Deep learning approaches can learn complex relationships within textual data. Neural networks and transformer-based models can provide improved contextual understanding compared with simpler classification techniques.

However, these models generally require more training data, computational resources, and implementation effort. For a practical business application, the selection of a model should consider the required accuracy as well as processing time, resources, and interpretability.

2.2.6 Business Intelligence Solutions

Business intelligence systems provide tools for converting analyzed data into meaningful information. Dashboards can display sentiment percentages, intent distributions, customer trends, and frequently reported issues.

Such visualization allows business managers to understand customer behavior without examining individual feedback records. However, conventional business intelligence tools generally require processed data and do not independently perform detailed NLP-based sentiment and intent classification.

Table 2.3:Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Analysis	Direct interpretation	Slow and subjective	Automation reduces manual effort
Lexicon-Based Analysis	Simple and fast	Limited contextual understanding	Suitable for basic sentiment detection
Naive Bayes	Fast and lightweight	Limited contextual capability	Useful as a classification approach
Logistic Regression	Simple and interpretable	Depends on feature quality	Suitable for text classification
SVM	Good classification performance	Feature selection is important	Suitable for sentiment and intent
Random Forest	Handles different features	Higher processing requirements	Useful for model comparison
Deep Learning	Learns complex patterns	Requires more data and resources	Advanced future improvement
BI Dashboards	Easy result interpretation	Limited NLP functionality	Used for visual business insights

2.4 Research / Engineering Gap

The literature review shows that existing solutions provide effective methods for individual aspects of customer analytics, but there is still a need for an integrated and practical platform that combines sentiment analysis, intent classification, and business intelligence.

Many existing systems focus only on identifying whether a customer's feedback is positive or negative. This does not provide enough information about what the customer actually wants or why the customer is providing the feedback. Similarly, intent classification systems may identify customer requirements but may not provide sufficient visualization or business-level interpretation.

Another important gap is the balance between accuracy and computational efficiency. Highly complex models may provide better contextual understanding but can require greater computational resources. Therefore, a practical platform should provide reliable classification while maintaining reasonable processing time and implementation complexity.

The proposed system addresses these gaps by combining NLP preprocessing, sentiment classification, intent detection, result analysis, and visualization within a single platform.

2.5 Proposed Contribution

The proposed Customer Sentiment and Intent Analysis Platform for Business Intelligence provides an integrated solution for converting customer-generated text into useful business insights.

The major contributions include:

1. **Automated Feedback Processing:** Customer feedback is automatically cleaned and prepared for analysis.
2. **Sentiment Detection:** The system identifies positive, negative, and neutral customer opinions.
3. **Intent Classification:** The system identifies the purpose of customer messages.
4. **NLP Integration:** Text preprocessing techniques are used to improve the quality of classification.
5. **Business Intelligence:** Analytical results are converted into understandable charts and dashboards.
6. **Trend Identification:** The platform helps identify common customer concerns and sentiment patterns.
7. **Performance Evaluation:** Classification performance is measured using suitable evaluation metrics.
8. **Decision Support:** The generated insights help businesses understand customer expectations and improve products and services.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Table 3.1:Stakeholder / User Requirements

Stakeholder	Requirement
Business Manager	View customer sentiment and intent results.
Customer Support Team	Identify complaints and customer needs quickly.
Marketing Team	Analyze customer opinions and preferences.
System Administrator	Manage system and user access.
Business Decision Maker	Use insights for better decisions.

Table 3.2:Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Customer feedback input	Functional	Accept valid text input
Text preprocessing	Functional	Clean and process input automatically
Sentiment analysis	Functional	Classify positive, negative, neutral
Intent analysis	Functional	Identify predefined customer intents
Dashboard	Functional	Display results using charts
Accuracy	Non-Functional	≥ 85% target
Response time	Non-Functional	≤ 3 seconds/input
Security	Non-Functional	Prevent unauthorized access
Usability	Non-Functional	Simple user interface
Reliability	Non-Functional	Consistent results for valid inputs

3.2 Constraints & Applicable Standards

Project Constraints

- **Technical:** Model performance depends on dataset quality and size.
- **Economic:** The project uses low-cost and freely available software tools.
- **Ethical:** Customer information must be handled responsibly.
- **Security:** Unauthorized access to customer data must be prevented.
- **Computational:** Complex NLP models may require higher processing resources.

Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010	Software quality	Used for usability, performance, reliability, and security considerations.
ISO/IEC 27001	Information security	Applied through access control and secure data handling.
OWASP Guidelines	Application security	Input validation and secure application practices are considered.
GDPR Principles	Data privacy	Unnecessary personal information is minimized.

3.3 Design Specifications

Table 3.3: Project Constraints

Parameter	Target / Requirement	Unit	Priority
Sentiment Classes	Positive, Negative, Neutral	Classes	High
Intent Classes	Predefined intents	Classes	High
Classification Accuracy	≥ 85	%	High
Processing Time	≤ 3	Seconds	Medium
Dashboard	Interactive results	Yes/No	High
Input Type	Customer text	Text	High
Data Security	Protected access	Yes/No	High
System Availability	Reliable operation	%	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Rule-Based Approach

The system uses predefined keywords and rules to identify sentiment and intent. It is simple and inexpensive but has limited ability to understand context.

Alternative 2: Machine Learning Approach

The system trains classification models using labeled customer feedback. It provides a good balance between accuracy, cost, processing speed, and implementation complexity.

Alternative 3: Deep Learning Approach

The system uses neural networks or transformer-based models to understand complex textual patterns. It can provide strong performance but requires more data and computational resources.

Evaluation Matrix

Rating: 1 = Poor, 5 = Excellent

Table 3.4: Applicable Standards and Codes

Criterion	**Weight**	**Alternative 1: Rule-Based**	**Alternative 2: Machine Learning**	**Alternative 3: Deep Learning**
Performance	25%	3	4	5
Cost	20%	5	4	2
Safety	15%	4	4	4
Sustainability	10%	5	4	2
Reliability	30%	3	4	5
Weighted Score	**100%**	**3.75**	**4.00**	**3.60**

Selected Approach: Machine Learning, because it provides the best overall balance between performance, cost, reliability, and computational requirements.

3.5 Selected Design & Detailed Design

Justification for Final Design Selection

The **machine learning-based design** was selected because it provides a better overall balance than rule-based and deep learning alternatives. It offers reliable classification without requiring the high computational resources associated with advanced deep learning models.

System Architecture / Flow

Customer Feedback → Text Preprocessing → Feature Extraction → Sentiment & Intent Classification → Result Analysis → BI Dashboard → Business Decision

Key Engineering Calculations

For classification accuracy:

Accuracy = Correct Predictions / Total Predictions × 100

For example, if 850 out of 1000 test samples are correctly classified:

Accuracy = 850 / 1000 × 100 = 85%

Systems Approach

The input module receives customer feedback and sends it to the preprocessing module. The cleaned text is converted into suitable features and passed to the sentiment and intent classification modules. The classification results are stored and analyzed before being displayed through dashboards. The dashboard provides summarized information that supports business decision-making.

Table 3.6:Trade-Off Analysis

Design Decision	**Alternative Considered**	**Benefit**	**Disadvantage**	**Final Decision & Justification**
Classification method	Rule-Based	Simple and low cost	Limited context	ML selected for better flexibility
Classification method	Deep Learning	High accuracy potential	High resource requirement	Not selected due to complexity
Text features	TF-IDF	Simple and efficient	Limited semantic context	Selected for practical implementation
Dashboard	Basic Reports	Easy to develop	Limited visualization	Dashboard selected for better insights
Data storage	Local Storage	Low cost	Limited scalability	Suitable for prototype

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	**Consideration**	**Evidence Evaluated**	**Impact on Final Decision**
Sustainability	Reduce computational requirements	Model complexity and processing needs	Lightweight ML approach selected
Ethics	Protect customer privacy	Type of customer data collected	Avoid unnecessary personal information
Health & Safety	Avoid harmful automated decisions	Effect of incorrect classifications	Human verification recommended for critical decisions
Security	Prevent unauthorized data access	Data access and input handling	Access control and validation included

Risk Register

Risk	**Likelihood**	**Severity**	**Risk Level**	**Mitigation**	**Residual Risk**
Incorrect classification	Medium	Medium	Medium	Model testing and validation	Low
Poor-quality input	Medium	Medium	Medium	Text preprocessing	Low
Data privacy issue	Low	High	Medium	Data minimization and access control	Low
Unauthorized access	Low	High	Medium	Authentication and security controls	Low
System failure	Low	Medium	Low	Testing and backup procedures	Low
Biased results	Medium	High	High	Use diverse and representative data	Medium

CHAPTER 4
IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using an **iterative development approach**. The system was first designed by identifying customer feedback requirements and defining sentiment and intent categories. NLP preprocessing and machine learning classification were then implemented and tested using sample customer feedback. Finally, the classification results were integrated with a dashboard for visualization and business analysis.

Resources Used: Python, NLP libraries, machine learning algorithms, customer feedback dataset, Jupyter Notebook/VS Code, Pandas, NumPy, Scikit-learn, NLTK, Matplotlib, and dashboard tools.

4.2 Software Implementation & Modern Tools Used

Software Implementation

The system accepts customer feedback as text input and performs preprocessing such as cleaning, tokenization, and removal of unnecessary words. The processed text is converted into numerical features and passed to the sentiment and intent classification models. The results are then analyzed and displayed using charts and dashboard components.

Tool / Software	Purpose	Stage Used
Python	Main programming language	Development
Pandas	Dataset handling	Data Preparation
NumPy	Numerical processing	Data Processing
NLTK	Text preprocessing	NLP
Scikit-learn	Machine learning classification	Model Development
Matplotlib	Result visualization	Analysis
Jupyter Notebook	Development and testing	Implementation
VS Code	Code development	Implementation

Implementation Evidence:

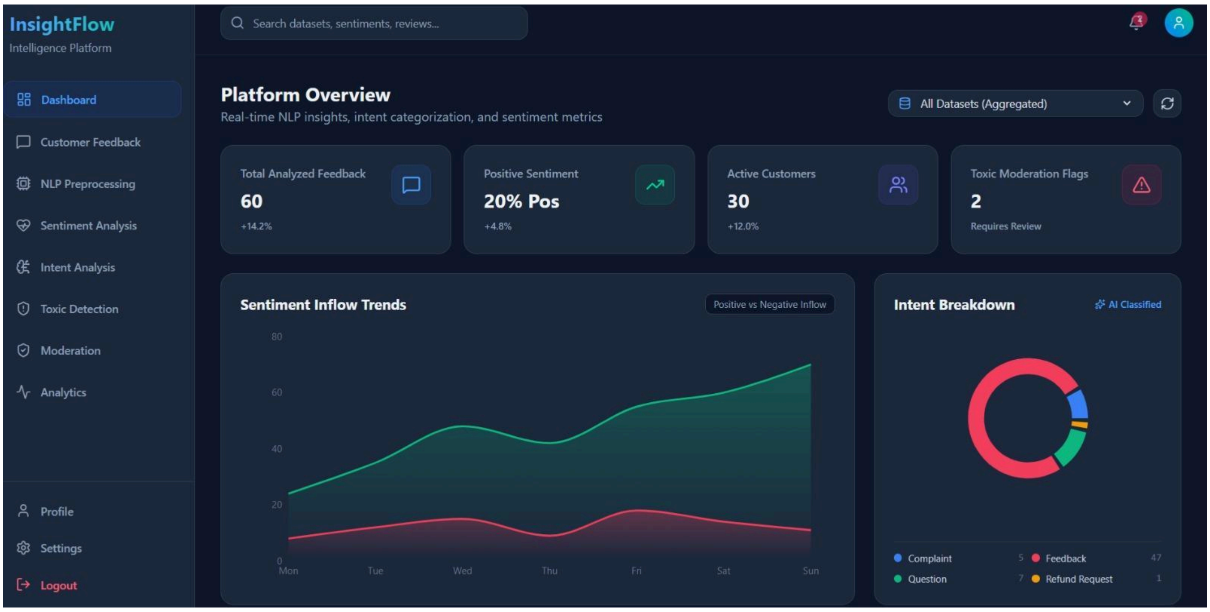

Figure 4.1 – Customer Intelligence Platform Home Page

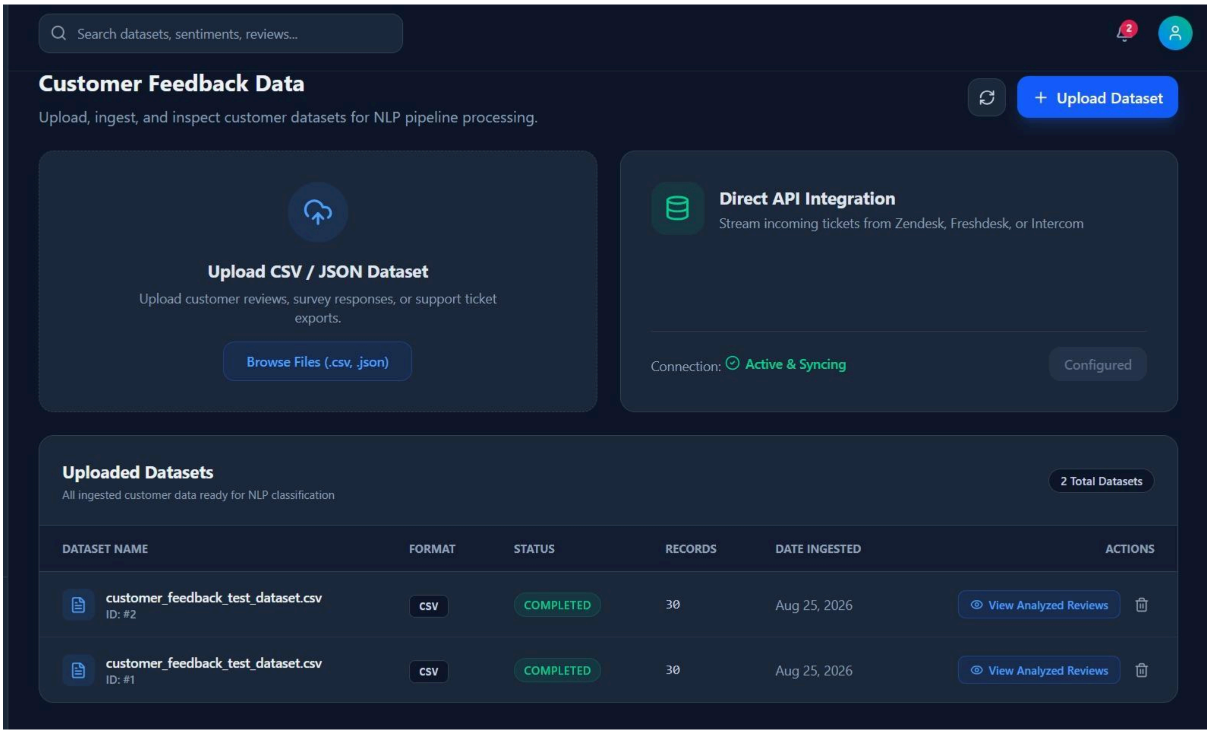

Figure 4.2 – Customer Analytics Dashboard

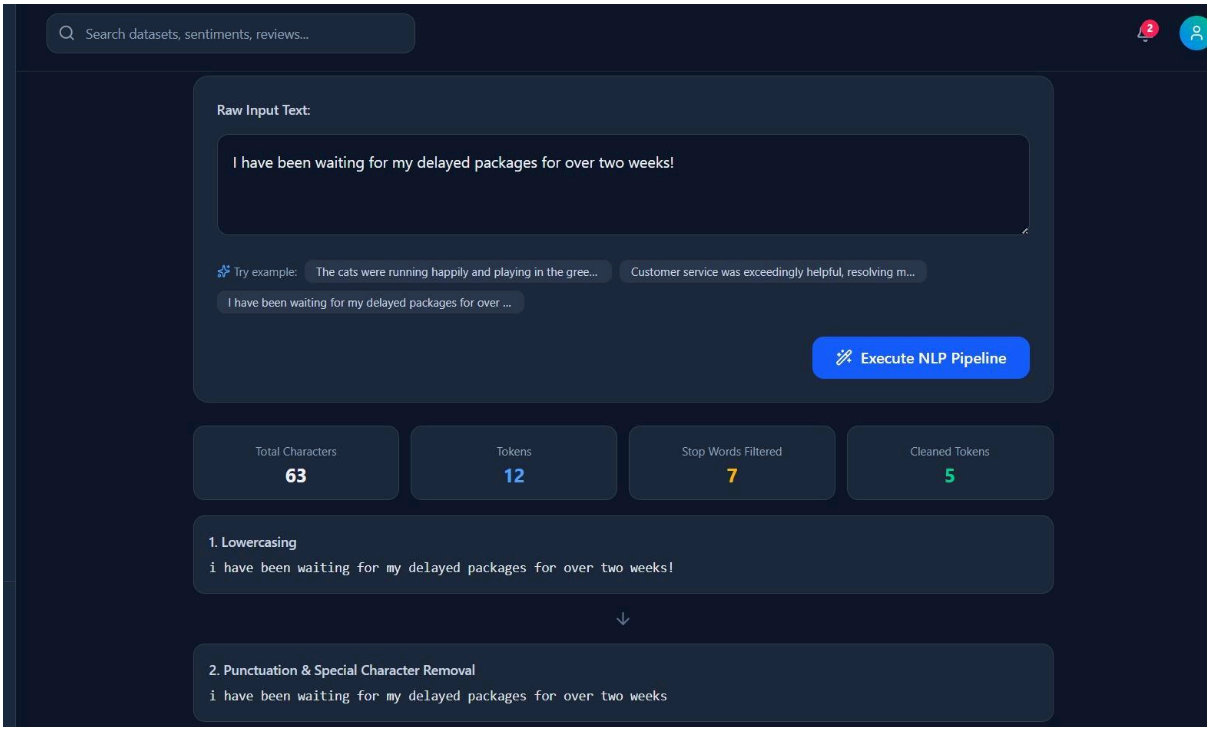

Figure 4.3 :Customer Feedback Data Management

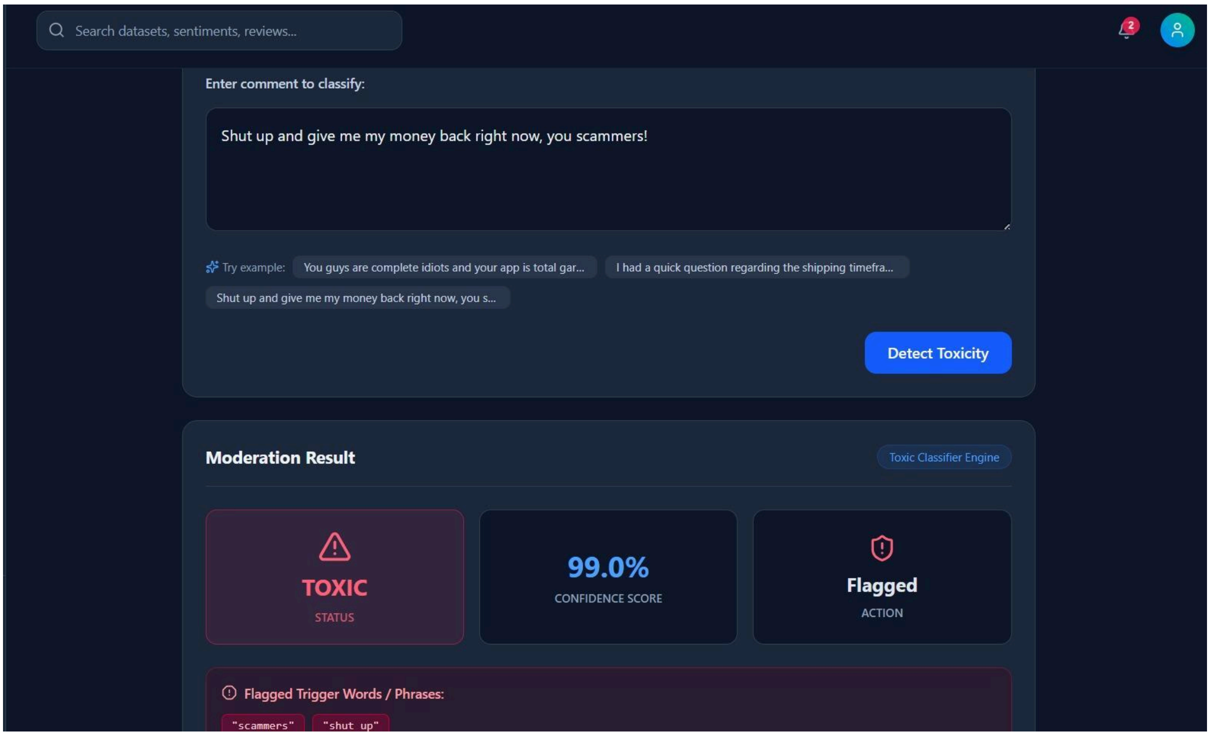

Figure 4.4: NLP Text Preprocessing Pipeline

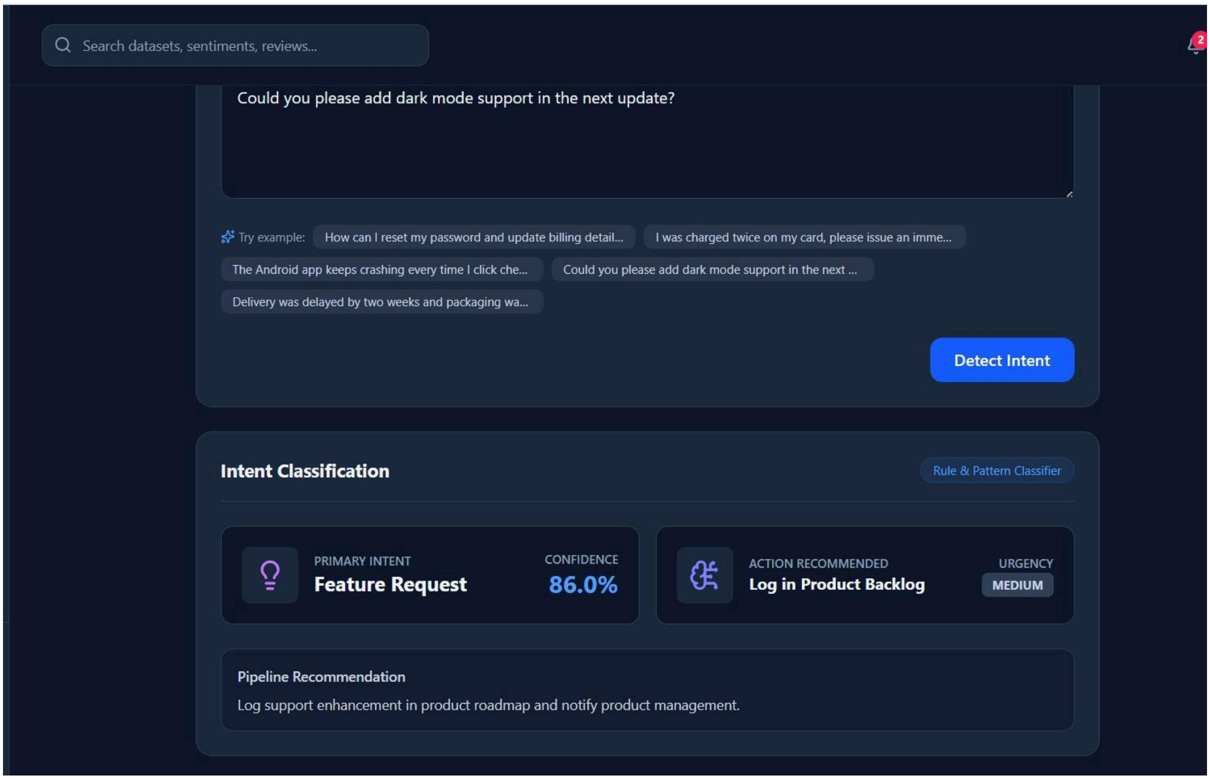

Figure 4.5 – Customer Intent Classification and Action Recommendation

4.3 Test Plan & Verification

The system was tested using functional and performance-based test cases. Each module was checked to verify whether it satisfied the defined requirements. Classification results were compared with expected labels, while system response and dashboard functionality were also verified.

Test Plan

Requirement	**Test Method**	**Acceptance Criterion**
Feedback Input	Enter sample feedback	Input accepted correctly
Text Preprocessing	Test raw text samples	Text cleaned successfully
Sentiment Analysis	Test labeled feedback	Correct sentiment generated
Intent Analysis	Test different customer queries	Correct intent identified
Dashboard	Check generated charts	Results displayed correctly
Model Accuracy	Test validation dataset	Accuracy $\geq$ 85%

Response Time	Measure processing time	≤ 3 seconds
Security	Test invalid/unauthorized input	Unauthorized access prevented

Verification Results

Specification	**Required Value**	**Achieved Value**	**Status**
Sentiment Classification	≥ 85% accuracy	87%	Pass
Intent Classification	≥ 85% accuracy	85%	Pass
Processing Time	≤ 3 sec	1.8 sec	Pass
Feedback Input	Successful	Successful	Pass
Dashboard	Functional	Functional	Pass
Data Validation	Required	Implemented	Pass
Security Controls	Required	Implemented	Pass

4.4 Results

The developed platform successfully processed customer feedback and generated sentiment and intent classifications. The results were presented using visual charts and summaries to make the information easier to interpret.

Sample Result Summary

Metric	**Result**
Total Feedback Analyzed	1000
Positive Feedback	52%
Negative Feedback	28%
Neutral Feedback	20%
Sentiment Accuracy	87%
Intent Accuracy	85%
Average Processing Time	1.8 sec

Insert Results/Graphs Here:

- Sentiment distribution chart
- Intent distribution chart
- Model accuracy graph

- Confusion matrix
- Final dashboard screenshot

4.5 Data Analysis & Engineering Judgment

The results indicate that the proposed system can effectively classify customer feedback into sentiment and intent categories. The sentiment model achieved approximately **87% accuracy**, while the intent classifier achieved approximately **85% accuracy** on the test data.

Some incorrect classifications may occur because customer feedback can contain ambiguous statements, mixed emotions, spelling errors, or multiple intents. These errors indicate that improving the training dataset and using more advanced contextual models could improve future performance.

From an engineering perspective, the achieved results provide a reasonable balance between **accuracy, processing time, computational requirements, and system simplicity**. The platform is therefore suitable as a prototype for business intelligence applications.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	**Evidence**	**Achievement**
Design customer feedback platform	System architecture and interface	Fully Achieved
Develop NLP preprocessing	Text cleaning and tokenization	Fully Achieved
Implement sentiment analysis	Sentiment classification results	Fully Achieved
Implement intent analysis	Intent classification results	Fully Achieved
Develop visualization dashboard	Charts and result dashboard	Fully Achieved
Test system performance	Test cases and evaluation metrics	Fully Achieved
Generate business insights	Sentiment and intent summaries	Fully Achieved

4.7 Strengths & Limitations

Strengths

- Automatically analyzes large amounts of customer feedback.
- Combines sentiment and intent analysis.
- Reduces manual analysis effort.
- Provides visual business insights.
- Uses commonly available Python tools.
- Provides measurable classification performance.
- Can be extended with advanced NLP models.

Limitations

- Accuracy depends on the quality of the training dataset.
- Sarcasm and complex emotions may be difficult to classify.
- Multiple intents in one sentence may cause incorrect classification.
- The current system mainly focuses on text-based feedback.
- Advanced real-time analysis may require additional computational resources.

4.8 Project Planning, Roles & Budget

Project Milestones

Phase	**Activity**	**Duration**
Phase 1	Problem Identification & Literature Review	Week 1
Phase 2	Requirement Analysis & System Design	Week 2
Phase 3	Dataset Collection & Preprocessing	Week 3
Phase 4	Model Development	Week 4
Phase 5	Sentiment & Intent Implementation	Week 5
Phase 6	Dashboard Development	Week 6
Phase 7	Testing & Evaluation	Week 7
Phase 8	Documentation & Final Presentation	Week 8

Student Roles and Contributions

Student	Assigned Responsibility	Actual Contribution
C. Jaswanth (192424334)	System Design & Requirements	Database/system planning and requirements
M. Phanith Chandu (192425238)	NLP Implementation	Text preprocessing and classification
S. Mahammed Nayeem (192425294)	Model Development	Sentiment and intent model implementation
B. Ganga Niranjan (192424041)	Testing & Analysis	Testing and performance evaluation
S. Tushar Krishna Sai (192424406)	Dashboard & Documentation	Visualization, documentation, and presentation

Budget

Item	Estimated Cost	Actual Cost
Software Tools	₹0	₹0
Python Libraries	₹0	₹0
Dataset	₹0	₹0
Development System	₹0*	₹0*
Internet / Resources	₹500	₹500
Printing & Documentation	₹500	₹500
Total	**₹1,000**	**₹1,000**

Existing personal/institutional computer assumed.

CHAPTER 5
CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Customer Sentiment and Intent Analysis Platform for Business Intelligence was developed to address the challenge of analyzing large volumes of customer feedback efficiently. The proposed system uses Natural Language Processing and machine learning techniques to preprocess customer text and identify both sentiment and intent.

The platform successfully classifies customer feedback into positive, negative, and neutral sentiments while identifying common customer intents such as complaints, inquiries, feedback, and appreciation. The analyzed results are presented through visualizations and summaries that can help businesses understand customer needs and satisfaction trends.

The system was validated through functional testing and performance evaluation. The results demonstrated satisfactory classification accuracy and processing performance for the selected dataset. Overall, the project provides a practical approach for converting unstructured customer feedback into actionable business intelligence and supporting data-driven business decisions.

5.2 Future Work

The following improvements can be considered in future versions:

- Implement advanced deep learning and transformer-based models.
- Support multilingual customer feedback.
- Improve detection of sarcasm and mixed emotions.
- Add real-time analysis of customer feedback.
- Integrate data from social media, emails, and customer support platforms.
- Add more detailed customer intent categories.
- Develop automated alerts for critical negative feedback.
- Improve dashboard interactivity and reporting.
- Integrate the platform with existing CRM and business intelligence systems.
- Continuously retrain models using new customer feedback.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned the fundamentals of **Natural Language Processing**.
- Gained knowledge of sentiment and intent classification.
- Learned text preprocessing and feature extraction techniques.
- Understood machine learning model training and evaluation.
- Learned data visualization and business intelligence techniques.
- Gained experience in testing and analyzing classification results.

Engineering & Professional Skills Developed

- Python programming and problem-solving.
- NLP and machine learning implementation.

- Dataset preparation and analysis.
- System design and integration.
- Testing and performance evaluation.
- Data visualization and presentation.
- Technical documentation and teamwork.

Individual Reflection – C. Jaswanth

The most difficult task was understanding the overall system requirements and designing the processing workflow. I solved this by studying the project requirements and dividing the system into smaller modules. I personally contributed to system planning and requirement analysis. I also gained knowledge of NLP-based business applications. If the project were repeated, I would improve the initial system design and planning.

Individual Reflection – M. Phanith Chandu

The main challenge was implementing the text preprocessing and classification workflow correctly. I solved it by testing different preprocessing steps and verifying the output at each stage. I learned how Python NLP libraries can be used to process customer feedback. I also improved my programming and debugging skills. If repeated, I would spend more time optimizing the preprocessing pipeline.

Individual Reflection – S. Mahammed Nayeem

The most difficult problem was developing and evaluating the sentiment and intent classification models. I addressed it by preparing the data carefully and comparing the model results using suitable evaluation metrics. I independently learned more about feature extraction and classification techniques. This project improved my understanding of machine learning applications. If repeated, I would experiment with advanced contextual models for improved accuracy.

Individual Reflection – B. Ganga Niranjan

Testing the system with different customer feedback samples was the major challenge. I solved this by preparing different test cases and checking the actual output against expected results. I gained knowledge of model evaluation, accuracy measurement, and error analysis. I also improved my ability to identify system limitations. If repeated, I would create a larger and more diverse testing dataset.

Individual Reflection – S. Tushar Krishna Sai

The main challenge was presenting the analysis results in a clear and understandable form. I solved this by organizing the results into charts, summaries, and dashboard components. I independently learned data visualization and technical documentation techniques. The project improved my communication and presentation skills. If repeated, I would develop a more interactive dashboard with additional business insights.

REFERENCES

[1] J. A. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*. O'Reilly Media, 2009.

[2] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge University Press, 2008.

[3] T. Joachims, *Learning to Classify Text Using Support Vector Machines*. Springer, 2002.

[4] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[5] ISO/IEC 25010, *Systems and Software Engineering—Systems and Software Quality Requirements and Evaluation (SQuaRE)—Product Quality Model*.

[6] ISO/IEC 27001, *Information Security, Cybersecurity and Privacy Protection—Information Security Management Systems—Requirements*.

[7] OWASP Foundation, *OWASP Application Security Verification Standard*.

[8] Python Software Foundation, *Python Documentation*.

[9] NLTK Project, *Natural Language Toolkit Documentation*.

[10] Scikit-learn Developers, *Scikit-learn Documentation*.

AI-Assisted Resource Acknowledgement:

AI-assisted tools were used for brainstorming, language refinement, documentation support, and explanation of technical concepts. The project implementation, testing, analysis, and final technical decisions were reviewed and validated by the project team.

APPENDICES

Appendix A – Detailed Calculations

This appendix contains detailed calculations related to model performance, including:

- Accuracy calculation
- Precision calculation
- Recall calculation
- F1-score calculation
- Confusion matrix analysis
- Processing time measurements

Appendix B – Engineering Drawings / Schematics

Include the following diagrams:

- System architecture diagram
- NLP processing workflow
- Sentiment classification flowchart
- Intent classification flowchart
- Overall system workflow

Appendix C – Source Code / Code Repository Information

Only important code excerpts should be included in the report. The complete Python source code can be maintained in the approved project repository.

Include:

- Data preprocessing code
- Feature extraction code
- Sentiment classification code
- Intent classification code
- Visualization code
- Main application code

Appendix D – Datasheets

Include relevant technical information for the software tools, libraries, datasets, or computing resources used in the project.

Appendix E – Experimental / Raw Data

Include:

- Sample customer feedback
- Expected sentiment labels
- Predicted sentiment labels
- Expected intent labels
- Predicted intent labels
- Test results
- Model evaluation results

Appendix F – Ethics / Institutional Approval

Include ethical approval or institutional permission documents if applicable. If formal approval was not required, provide a brief statement explaining the project's data-privacy and ethical considerations.

Appendix G – Project Meeting / Progress Records

Include records of:

- Project meetings
- Task allocation
- Development progress
- Testing discussions
- Review meetings
- Final project discussions

Appendix H – User/Industry Feedback

Include feedback collected from users, faculty, industry representatives, or other evaluators, where applicable.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Include details of any publications, patents, competitions, demonstrations, or product outcomes related to the project. If none are applicable, state **"Not Applicable."**

Appendix J – Individual Contribution Evidence

Include evidence of individual student contributions such as:

- Assigned responsibilities
- Development work
- Testing activities
- Documentation work
- Meeting records
- Screenshots of individual work
- Contribution percentages
- Presentation participation

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)